Huikai Shao

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Experience

- **Xi'an Jiaotong University** *Jul. 2026 -*
  Associate Professor Xi'an, China
- **Xi'an Jiaotong University** *Sept. 2022 - Jul. 2026*
  Assistant Professor, Postdoctor Xi'an, China

Education

- **Xi'an Jiaotong University** *Sept. 2017 - Dec. 2021*
  PhD, Control Science and Engineering Xi'an, China
- **Chongqing University** *Sept. 2013 - Jun. 2017*
  Bachelor, Automation Chongqing, China
- **University of Twente** *Nov. 2019 - Jan. 2020*
  Visiting student Enschede, The Netherlands

Research activities

- **Biometrics, Palmprint recognition, Palm vein recognition, Dorsal hand vein recognition**
- Research on cross-device palmprint recognition methods based on domain adaptation and domain generalization, NSFC.
- Development and Application of Localized High Security Authentication System Based on Palmprint and Palm Vein Recognition, Key Research and Development Program of Shaanxi.
- Palmprint recognition system, Xi'an Science and Technology Project.

Academic and professional reputation

- Member of the editorial board of the journal Discover Computing, 2025-.
- Workshop Chair, ISAICS, 2025.
- Leader Guest Editor, Electronics, 2024.
- Young Talent Fund of Association for Science and Technology in Shaanxi, China, 2022.
- Deputy Secretary-General of the Biometric Recognition Technical Committee of CSIG, 2025-
- Reviewer for IEEE TIP, IEEE TIFS, IEEE TMM, IEEE TIM, IEEE TR, PR, AAAI, ...

Publications

[1] **H. Shao**, Y. Zou, C. Liu, Q. Guo, and D. Zhong, " Learning to Generalize Unseen Dataset for Cross-Dataset Palmprint Recognition," IEEE Transactions on Information Forensics and Security (TIFS), vol. 19, pp. 3788-3799, 2024.

[2] **H. Shao**, C. Liu, X. Li and D. Zhong, "Privacy Preserving Palmprint Recognition via Federated Metric Learning," IEEE Transactions on Information Forensics and Security (TIFS), vol. 19, pp. 878-891, 2024.

[3] **H. Shao**, and D. Zhong, "Learning with Partners to Improve the Multi-source Cross-dataset Palmprint Recognition," IEEE Transactions on Information Forensics and Security (TIFS), vol. 16, pp. 5182-5194, 2021.

[4] **H. Shao**, S. Shi, X. Du, D. Zeng, and D. Zhong, "Robust palmprint recognition via multi-stage noisy label selection and correction," IEEE Transactions on Image Processing (TIP), vol. 34, pp. 4591-4601, 2025.

[5] Y. Zou, **H. Shao***, C. Liu, S. Zhu, Z. Hou, and D. Zhong, "PFIG-Palm: Controllable Palmprint Generation via Pixel and Feature Identity Guidance," IEEE Transactions on Image Processing (TIP), vol. 34, pp. 4591-4601, 2025.

[6] **H. Shao**, P. Li and D. Zhong, "Generating Stylized Features for Single-Source Cross-Dataset Palmprint Recognition with Unseen Target Dataset," IEEE Transactions on Image Processing (TIP), vol. 33, pp. 4911-4922, 2024.

[7] **H. Shao**, and D. Zhong, "Towards Cross-Dataset Palmprint Recognition Via Joint Pixel and Feature Alignment," IEEE Transactions on Image Processing (TIP), vol. 30, pp. 3764-3777, 2021.

[8] Y. Zou, **H. Shao***, L. Fang, Z. Xiong, and D. Zhong*, "FlowPalm: Optical Flow Driven Non Rigid Deformation for Geometrically Diverse Palmprint Generation," in 2026 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[9] **H. Shao**, D. Jing, N. Xia, Z. Tang, and D. Zhong, "Universal domain adaptation in intelligent fault diagnosis via simulating unseen classes," IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 9178-9189, 2024.

[10] D. Jing, **H. Shao*** and D. Zhong, "Boosting Edge Detection via Fusing Spatial and Frequency Domains," Pattern Recognition, vol. 155, pp. 1-12, 2024.

[11] Y. Zou, C. Liu, **H. Shao*** and D. Zhong, "Unsupervised Palmprint Image Quality Assessment via Pseudo-Label Generation and Ranking Guidance," IEEE Transactions on Instrumentation and Measurement (TIM), vol. 72, pp. 1-11, 2023.

[12] **H. Shao**, and D. Zhong, "Multi-target Cross-dataset Palmprint Recognition via Distilling From Multi-Teachers," IEEE Transactions on Instrumentation and Measurement (TIM), vol. 72, pp. 1-14, 2023.

[13] **H. Shao**, and D. Zhong, "Towards Open-set Touchless Palmprint Recognition via Weight-based Meta Metric Learning," Pattern Recognition, vol. 121, pp. 1-12, 2022.

[14] **H. Shao**, D. Zhong, and X. Du, "Deep Distillation Hashing for Unconstrained Palmprint Recognition," IEEE Transactions on Instrumentation and Measurement (TIM), vol. 70, pp. 1-13, 2021.

[15] **H. Shao**, D. Zhong, X. Du, S. Du, and R. Veldhuis, "Few-Shot Learning for Palmprint Recognition via Meta-Siamese Network," IEEE Transactions on Instrumentation and Measurement (TIM), vol. 70, pp. 1-12, 2021.

[16] **H. Shao**, and D. Zhong, "One-shot cross-dataset palmprint recognition via adversarial domain adaptation," Neurocomputing, vol. 432, pp. 288-299, 2021.

[17] **H. Shao**, D. Zhong, and X. Du, "A Deep Biometric Hash Learning Framework for Three Advanced Hand-based Biometrics," IET Biometrics, vol. 10, no. 3, pp. 246-259, 2021.

[18] **H. Shao**, D. Zhong, and X. Du, "Effective deep ensemble hashing for open-set palmprint recognition," Journal of Electronic Imaging, vol. 29, no. 01, pp. 1–19, 2020.

[19] **H. Shao**, and D. Zhong, "Towards privacy palmprint recognition via federated hash learning," Electronics Letters, vol. 56, no. 25, pp. 1418-1420, 2020.

[20] **H. Shao**, and D. Zhong, "Few-shot palmprint recognition via Graph Neural Networks," Electronics Letters, vol. 55, no. 16, pp. 890-892, 2019.

[21] **H. Shao**, D. Zhong, and X. Du, "Efficient Deep Palmprint Recognition via Distilled Hashing Coding," in 2019 IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshop, 2019.

[22] **H. Shao**, D. Zhong, and Y. Li, "PALMGAN for Cross-domain Palmprint Recognition," in 2019 IEEE International Conference on Multimedia Expo (ICME), 2019.

[23] **H. Shao**, D. Zhong, and X. Du, "Cross-domain Palmprint Recognition Based on Transfer Convolutional Autoencoder," in 2019 IEEE International Conference on Image Processing (ICIP), 2019.

[24] Q. Guo, **H. Shao***, C. Liu, J. Wan, D. Zhong, "Homomorphic Encryption-Based Privacy Protection for Palmprint Recognition", in Chinese Conference on Biometric Recognition, 2023

[25] D. Zhong, **H. Shao**, and X. Du, "A Hand-based Multi-biometrics via Deep Hashing Network and Biometric Graph Matching," IEEE Transactions on Information Forensics and Security (TIFS), vol. 14, pp. 3140-3150, 2019.

[26] C. Liu, D. Zhong, and **H. Shao**, "Data Protection in Palmprint Recognition via Dynamic Random Invisible Watermark Embedding," IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), vol. 32, no. 10, pp. 6927-6940, 2022

[27] J. Wan, D. Zhong, and **H. Shao**, "Palmprint recognition system for mobile device based on circle loss," Displays, vol. 73, pp. 1–9, 2022.

[28] X. Du, D. Zhong, and **H. Shao**, "Cross-Domain Palmprint Recognition Via Regularized Adversarial Domain Adaptive Hashing," IEEE Transactions on Circuits and Systems for Video Technology (TCSVT), vol. 31, no. 6, pp. 2372-2385, 2021.

[29] C. Liu, D. Zhong, and **H. Shao**, "Few-shot Palmprint recognition based on Similarity Metric Hashing Network," Neurocomputing, vol. 456, pp. 540-549, 2021.

[30] D. Zhong, **H. Shao**, and S. Liu, "Towards application of hand dorsal vein recognition under uncontrolled environment based on biometric graph matching," IET Biometrics, vol. 8, no. 2, pp. 159-167, 2019.